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SEMI E90.1-1224
SPECIFICATION FOR SECS-II PROTOCOL SUBSTRATE TRACKING

This Standard was technically approved by the Information & Control Global Technical Committee. This edition was approved for publication by the global Audits and Reviews Subcommittee on December 10, 2024. Available at www.semiviews.org and www.semi.org in December 2024; originally published March 2001; previously published June 2024.

1 Purpose

1.1 This Specification maps the services and data of SEMI E90 to SECS-II streams and functions, and data definitions.

2 Scope

2.1 This Specification covers equipment supporting automated substrate tracking.

2.2 This Specification applies to all implementations of SEMI E90 that use the SECS-II message protocol (SEMI E5). Compliance to this Standard requires compliance to both SEMI E90 and SEMI E5.

NOTICE: SEMI Standards and Safety Guidelines do not purport to address all safety issues associated with their use. It is the responsibility of the users of the Documents to establish appropriate safety and health practices, and determine the applicability of regulatory or other limitations prior to use.

3 Referenced Standards and Documents

3.1 *SEMI Standards and Safety Guidelines*

SEMI E5 — Specification for SEMI Equipment Communications Standard 2 Message Content (SECS-II)

SEMI E39.1 — Specification for SECS-II Protocol for Object Services (OSS)

SEMI E90 — Specification for Substrate Tracking

NOTICE: Unless otherwise indicated, all documents cited shall be the latest published versions.

4 Mapping

4.1 This section shows the specific SECS-II streams and functions that shall be used for SECS-II implementation of the services defined in SEMI E90, as well as the parameter mapping for data attached to services.

4.2 *Message Mapping*

4.2.1 *Services Message Mapping* — Table 1 defines the relationships between SEMI E90 services and SECS-II messages.

4.2.2 *Event Message Mapping* — Table 2 defines the relationships between SEMI E90 collection events and SECS-II messages.

Table 1 Services Message Mapping Table

Service Name	*Stream, Function*	*SECS-II Message Name*
registers substrate	S14,F9/10	Create Object Request/Acknowledge (SEMI E39.1)
remove substrate	S14,F11/12	Delete Object Request/Acknowledge (SEMI E39.1)
UpdateSubstrateObject	S14F19/F20	Generic Service Request/Acknowledge
ProceedWithSubstrate	S14F19/F20	Generic Service Request/Acknowledge
CancelSubstrate	S14F19/F20	Generic Service Request/Acknowledge

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Table 2 Event Message Mapping Table

Event Name	*Stream, Function*	*SECS-II Message Name*
All state model transitions	S6F11/12	Event Report Send/Acknowledge

4.3 *Parameter Mapping* — Table 3 defines the relationships between SEMI E90 service parameters and SECS-II data definitions.

Table 3 Service Parameters to SECS-II Data Items Mapping

Parameter Name	*Range*	*SECS-II Data Item*
AdditionalInfo	List of structure with name-value pairs.	L,n 1. L,2 1. $AdditionalInfoName_1$ 2. $AdditionalInfoValue_1$: n. L,2 1. $AdditionalInfoName_n$ 2. $AdditionalInfoValue_n$ $AdditionalInfoName_i$ is Format Code 20, and is limited to 1–256 characters. $AdditionalInfoValue_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
ErrorCode	Enumerated	ERRCODE
ErrorText	1 to 80 characters	ERRTEXT
ObjectAction	‘UpdateSubstrateObject’	SVCNAME
ObjectActionStatus	List of service acknowledge and status.	L,2 1. SVCACK 2. Status
ObjectAction ParameterResult	(List of) name value pairs providing information about the attributes successfully set.	L,n 1. L,2 1. $SPNAME_1$ 2. $SPVAL_1$: n. L,2 1. $SPNAME_n$ 2. $SPVAL_n$
ObjectLinkID	51	LINKID
OperationID	51	OPID
STAcknowledge	0 if success; 1 if error (Conforms to ObjAck in SEMI E39.1)	OBJACK
Status	List of error information.	L,n n = number of errors 1. L,2 1. <$ErrorCode_1$> 2. <$ErrorText_1$> . . n. L,2 1. <$ErrorCode_n$> 2. <$ErrorText_n$>

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Parameter Name	*Range*	*SECS-II Data Item*
STStatus	STAcknowledge and Status.	L,2 1. <STAcknowledge> 2. <Status>
SubstID	1–80 characters (Conforms to ObjID in SEMI E39.1)	OBJID
SubstIDList (ordered list of)	List of Substrate IDs.	L,n n = number of substrate identifiers 1. <$SubstID_1$> . . n. <$SubstID_n$>
SubstLocID	1–80 characters (Conforms to ObjID in SEMI E39.1)	OBJID
SubstLocIDList (ordered list of)	List of Substrate Location ID's corresponding to the SubstIDList (ordered list of).	L,n n = number of substrate identifiers in SubstIDList (ordered list of) 1. <$SubstLocID_1$> . . n. <$SubstLocID_n$>
SubstProcState	Enumerated per Variable SubstProcState	U1
SubstProcStateList (ordered list of)	List of Substrate Processing States corresponding to the SubstIDList (ordered list of).	L,n n = number of substrate identifiers in SubstIDList (ordered list of) 1. <$SubstProcState_1$> . . n. <$SubstProcState_n$>
SubstrateObject-Attributes	List of substrate object attribute names and values.	L,n 1. L,2 1. $SPNAME_1$ 2. $SPVAL_1$: n. L,2 1. $SPNAME_n$ 2. $SPVAL_n$
SubstState	Enumerated per Variable SubstState	U1
SubstStateList (ordered list of)	List of Substrate Transport States corresponding to the SubstIDList (ordered list of).	L,n n = number of substrate identifers in SubstIDList (ordered list of) 1. <$SubstState_1$> . . n. <$SubstState_n$>

4.4 *SECS-II Data Items Without Corresponding SEMI E90 Parameters* — Table 4 contains the SECS-II data items that do not correspond to SEMI E90's service parameter.

Table 4 Additional Data Item Requirements Table

Function	*SECS-II Data Item*
TimeIn from SubstHistory variable data definition	TIMESTAMP
TimeOut from SubstHistory variable data definition	A[0] if the substrate(s) remain in the location. TIMESTAMP if the substrate(s) have departed the location.

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5 Variable Data Item Mapping

5.1 Table 5 shows the specific SECS-II data classes, and formats needed for SECS-II implementations of SEMI E90 variable data items.

Table 5 Variable Data Item Mapping

Variable Name	*Class*	*Format*
AcquiredID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
AcquiredIDList	DVVAL	L,n 1. <$AcquiredID_1$> . . n. <$AcquiredID_n$> n = the number of substrates involved in the related state model transition. $AcquiredID_i$ has the same format as variable AcquiredID listed in this table.
AdditionalInfo	DVVAL	L,n n = number of substrate additional information name-value pairs 1. L,2 1. $AdditionalInfoName_1$ 2. $AdditionalInfoValue_1$: n. L,2 1. $AdditionalInfoName_n$ 2. $AdditionalInfoValue_n$ $AdditionalInfoName_i$ is Format Code 20, and is limited to 1–256 characters. $AdditionalInfoValue_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
AdditionalInfoList	DVVAL	L,m m = the number of substrates involved in the related state model transition. 1. <$AdditionalInfo_1$> . . n. < $AdditionalInfo_m$> $AdditionalInfo_i$ has the same format as variable AdditionalInfo listed in this table.
BatchLocDisableEvents	DVVAL	11 (Boolean)
$BatchLocDisableEvents_i$	SV	11 (Boolean)
BatchLocID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
$BatchLocID_i$	SV	A[1..80] (Conforms to ObjID in SEMI E39.1)
BatchLocObjType	DVVAL	A[1..40] (Conforms to ObjType in SEMI E39.1)
$BatchLocObjType_i$	SV	A[1..40] (Conforms to ObjType in SEMI E39.1)
BatchLocState	DVVAL	51(U1) Enumerated as: 0-UNOCCUPIED 1-OCCUPIED
$BatchLocState_i$	SV	51(U1) Enumerated as: 0-UNOCCUPIED 1-OCCUPIED

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Variable Name	*Class*	*Format*
BatchSubstIDMap	DVVAL	L,n n = number of positions in a batch 1. <SubstID$_1$> . . n. <SubstID$_n$>
BatchSubstIDMap$_i$	SV	L,n n = number of positions in a batch 1. <SubstID$_1$> . . n. <SubstID$_n$>
SubstBatchLocID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstBatchLocIDList	DVVAL	L,n 1. <SubstBatchLocID$_1$> . . n. <SubstBatchLocID$_n$> n = the number of substrates involved in the related state model transition. SubstBatchLocID$_i$ has the same format as variable SubstBatchLocID listed in this table.
SubstDestination	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstDestinationList	DVVAL	L,n 1. <SubstDestination$_1$> . . n. <SubstDestination$_n$> n = the number of substrates involved in the related state model transition. SubstDestination$_i$ has the same format as variable SubstDestination listed in this table.
SubstHistory	DVVAL	L,n n = number of locations visited by the substrate 1. L,3 1. <SubstLocID$_1$ or BatchLocID$_1$+"."+SubstPosInBatch$_1$> 2. <TimeIn$_1$> 3. <TimeOut$_1$> . . n. L,3 1. <SubstLocID$_n$ or BatchLocID$_n$ +"."+SubstPosInBatch$_n$ > 2. <TimeIn$_n$> 3. <TimeOut$_n$>
SubstHistoryList	DVVAL	L,n 1. <SubstHistory$_1$> . . n. <SubstHistory$_n$> n = the number of substrates involved in the related state model transition. SubstHistory$_i$ has the same format as variable SubstHistory listed in this table.
SubstID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)

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Variable Name	*Class*	*Format*
SubstIDList	DVVAL	L,n 1. <$SubstID_1$> . . n. <$SubstID_n$> n = the number of substrates involved in the related state model transition. $SubstID_i$ has the same format as variable SubstID listed in this table.
SubstIDStatus	DVVAL	51 (U1) Enumerated as: 0 – NOT CONFIRMED 1 – WAITING FOR HOST 2 – CONFIRMED 3 – CONFIRMATION FAILED
SubstIDStatusList	DVVAL	L,n 1. <$SubstIDStatus_1$> . . n. <$SubstIDStatus_n$> n = the number of substrates involved in the related state model transition. $SubstIDStatus_i$ has the same format as variable SubstIDStatus listed in this table.
SubstLocDisableEvents	DVVAL	11 (Boolean)
$SubstLocDisableEvents_i$	SV	11 (Boolean)
SubstLocID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstLocIDList	DVVAL	L,n 1. <$SubstSubstLocID_1$> . . n. <$SubstSubstLocID_n$> n = the number of substrates involved in the related state model transition. $SubstSubstLocID_i$ has the same format as variable SubstSubstLocID listed in this table.
$SubstLocID_i$	SV	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstLocObjType	DVVAL	A[1..40] (Conforms to ObjType in SEMI E39.1)
$SubstLocObjType_i$	SV	A[1..40] (Conforms to ObjType in SEMI E39.1)
SubstLocState	DVVAL	51(U1) Enumerated as: 0 – UNOCCUPIED 1 – OCCUPIED
$SubstLocState_i$	SV	51(U1) Enumerated as: 0 – UNOCCUPIED 1 – OCCUPIED
SubstLocSubstID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
$SubstLocSubstrID_i$	SV	A[1..80] (Conforms to ObjID in SEMI E39.1) This variable maps to status variable $SubstrLocSubstrID_i$ defined in SEMI E90.
SubstLotID	DVVAL	A[1..80]

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Variable Name	*Class*	*Format*
SubstLotIDList	DVVAL	L,n 1. <SubstLotID$_1$> . . n. <SubstLotID$_n$> n = the number of substrates involved in the related state model transition. SubstLotID$_i$ has the same format as variable SubstLotID listed in this table.
SubstMtrlStatus	DVVAL	U1 (equipment dependent enumeration)
SubstMtrlStatusList	DVVAL	L,n 1. <SubstMtrlStatus$_1$> . . n. <SubstMtrlStatus$_n$> n = the number of substrates involved in the related state model transition. SubstMtrlStatus$_i$ has the same format as variable SubstMtrlStatus listed in this table.
SubstObjType	DVVAL	A[1..40] (Conforms to ObjType in SEMI E39.1)
SubstObjTypeList	DVVAL	L,n 1. <SubstObjType$_1$> . . n. <SubstObjType$_n$> n = the number of substrates involved in the related state model transition. SubstObjType$_i$ has the same format as variable SubstObjType listed in this table.
SubstPosInBatch	DVVAL	A (range is equipment dependent)
SubstPosInBatchList	DVVAL	L,n 1. <SubstPosInBatch$_1$> . . n. <SubstPosInBatch$_n$> n = the number of substrates involved in the related state model transition. SubstPosInBatch$_i$ has the same format as variable SubstPosInBatch listed in this table.
SubstProcState	DVVAL	51(U1) Enumerated as: 0 – NEEDS PROCESSING 1 – IN PROCESS 2 – PROCESSED 3 – ABORTED 4 – STOPPED 5 – REJECTED 6 – LOST 7 – SKIPPED

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Variable Name	*Class*	*Format*
SubstProcStateList	DVVAL	L,n 1. <SubstProcState$_1$> . . n. <SubstProcState$_n$> n = the number of substrates involved in the related state model transition. SubstProcState$_i$ has the same format as variable SubstProcState listed in this table.
SubstrateReaderEnabled	EC	Binary 0 = Disabled 1 = Enabled
SubstSource	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstSourceList	DVVAL	L,n 1. <SubstSource$_1$> . . n. <SubstSource$_n$> n = the number of substrates involved in the related state model transition. SubstSource$_i$ has the same format as variable SubstSource listed in this table.
SubstState	DVVAL	51(U1) Enumerated as: 0 – AT SOURCE 1 – AT WORK 2 – AT DESTINATION
SubstStateList	DVVAL	L,n 1. <SubstState$_1$> . . n. <SubstState$_n$> n = the number of substrates involved in the related state model transition. SubstState$_i$ has the same format as variable SubstState listed in this table.
SubstSubstLocID	DVVAL	A[1..80] (Conforms to ObjID in SEMI E39.1)
SubstType	DVVAL	51(U1) Enumerated as: 0 – WAFER 1 – FLAT PANEL 2 – CD 3 – MASK
SubstTypeList	DVVAL	L,n 1. <SubstType$_1$> . . n. <SubstType$_n$> n = the number of substrates involved in the related state model transition. SubstType$_i$ has the same format as variable SubstType listed in this table.

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Variable Name	*Class*	*Format*
SubstUsage	DVVAL	51(U1) Enumerated as: 0 – PRODUCT 1 – TEST 2 – FILLER
SubstUsageList	DVVAL	L,n 1. <SubstUsage$_1$> . . n. <SubstUsage$_n$> n = the number of substrates involved in the related state model transition. SubstUsage$_i$ has the same format as variable SubstUsagelisted in this table.

6 SECS-II Attribute Definitions

6.1 *Substrate Object SECS-II Attributes Definitions* — The following are the SECS-II structure definitions for the SEMI E90 Substrate Object.

Table 6 Substrate Object SECS-II Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
AcquiredID	20(A) [0,80]
AdditionalInfo	L,n n = number of substrate additional information name-value pairs 1. L,2 1. AdditionalInfoName$_1$ 2. AdditionalInfoValue$_1$: n. L,2 1. AdditionalInfoName$_n$ 2. AdditionalInfoValue$_n$ AdditionalInfoName$_i$ is Format Code 20, and is limited to 1–256 characters. AdditionalInfoValue$_i$ is Format Code 0, 10, 11, 20, 3(), 4() or 5().
BatchLocID	<OBJID> BatchLocID
ObjType	‘Substrate’
ObjID	<OBJID> SubstID (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6)
LotID	20(A) LotID
MaterialStatus	51(U1) MaterialStatus MaterialStatus is an equipment specific enumerated value
SubstDestination	<OBJID> SubstLocID

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Attribute Name	*Attribute Data Form: SECS-II Structure*
SubstHistory	L,n n = number of locations visited by the substrate 1. L,3 1. <OBJID$_1$> SubstLocID$_1$ or BatchLocID$_1$+“.”+SubstPosInBatch$_1$ 2. <TIMESTAMP$_1$> TimeIn$_1$ 3. <TIMESTAMP$_1$ or A[0]$_1$> TimeOut$_1$ . . n. L,3 1. <OBJIDn> SubstLocID$_n$ or BatchLocID$_n$+“.”+SubstPosInBatch$_n$ 2. <TIMESTAMP$_n$> TimeIn$_n$ 3. <TIMESTAMP$_n$ or A[0]$_n$> TimeOut$_n$
SubstLocID	<OBJID> SubstLocID
SubstPosInBatch	20(A) SubstPosInBatch
SubstProcState	51(U1) SubstProcState SubstProcState enumerated per Variable SubstProcState.
SubstSource	<OBJID> SubstLocID
SubstState	51(U1) SubstState SubstState enumerated per Variable SubstState.
SubstType	51(U1) SubstType SubstType enumerated per Variable SubstType.
SubstUsage	51(U1) SubstUsage SubstUsage enumerated per Variable SubstUsage.
SubstIDStatus	51(U1) Enumerated 0 = NOT CONFIRMED 1 = WAITING FOR HOST 2 = CONFIRMED 3 = CONFIRMATION FAILED

6.2 *Substrate Location Object SECS-II Attributes Definitions* — The following are the SECS-II structure definitions for the SEMI E90 Substrate Location Object.

Table 7 Substrate Location Object Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
DisableEvents	11 (Boolean)
ObjType	‘SubstLoc’
ObjID	<OBJID> SubstLocID (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6)
SubstID	<OBJID> SubstID
SubstLocState	51(U1) SubstLocState SubstLocState enumerated per Variable SubstLocState.

6.3 *Batch Location Object SECS-II Attributes Definitions* — The following are the SECS-II structure definitions for the SEMI E90 Batch Location Object.

Table 8 Batch Location Object Attribute Definitions

Attribute Name	*Attribute Data Form: SECS-II Structure*
ObjID	<OBJID> BatchLocID (Conforms to the restrictions of ObjID as specified in SEMI E39.1, § 6)

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Attribute Name	*Attribute Data Form: SECS-II Structure*
ObjType	'BatchLoc'
BatchLocState	51(U1) BatchLocState BatchLocState enumerated per Variable BatchLocState.
BatchSubstIDMap	L,n n = number of positions in a batch 1. <OBJID$_1$> SubstID$_1$ . . n. <OBJID$_n$> SubstID$_n$
DisableEvents	11 (Boolean)

7 Related Documents

7.1 *SEMI Standards and Safety Guidelines*

SEMI E30 — Specification for the Generic Model for Communications and Control of Manufacturing Equipment (GEM)

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